

A control system has 8 devices (A-H). In the first time slot, only devices A, C, E, F, and H have data. In the second time slot, devices A, B, D, E, and G have data.

1. What TMD technique should be used here (synchronous or statistical), and why?
2. Sketch a figure of the TDM process and show the frames.
3. Sketch a figure of the other TDM process and show the frames.
4. Find out the frame size. [Hint: Look at the figure you drew: each frame has data slots + addressing bits (like a, b, ..., h, that are sent after the particular data slot A1, B1, ..., H1). Since there are 8 devices, how many addressing bits will you need?]

Solution to 4:

By drawing the figure in (2), we see that each frame has FIVE slots only.

Since unit is not given, let unit (or interleaved bits) = 8 bits

Now, we know that in Statistical TDM, each slot has TWO parts: one for the actual data like A1, A2, B1, C1, D2, etc (given by the unit, which in this case is 8), and addressing bits (like a for A, b for B, etc). Since there are 8 devices, we would need 3 addressing bits for each device (a: 000, b: 001, c: 010, d: 011, e: 100, f: 101, g: 110, h: 111).

So, frame size equation for statistical TDM = #slots * (interleaved bits or units + addressing bits)

[Remember, in synchronous the formula was: frame size = #slots * unit + synchronizing bits]

In this math,

Statistical TDM frame size = 5 * (8 + 3) = 55 bits

Synchronous TDM frame size = 8 * 8 + synchronizing bits = 64 + synchronizing bits [which is not given here]

Note: The actual answer to question 4 is the blue text only. The rest are only for explanation, not needed in the exam.